

Gop/Portfolio error:

```
gop_perimeter = {
    'IT': {'Gop': {'PDS'}, 'Perimeter':
{'ABD', 'BCD', 'KJH'}},
    'TS': {'Gop': {}, 'Perimeter': {'ABD', 'KJH'}},
    'RD': {'Gop': {'*All*'}, 'Perimeter': {'BCD', 'KJH'}}
}
gop_data = {
    "PDS": {"Perimeter": "BCD", "Status": "Active", "RDB" :
"1234", "Zone": "EUR"},
    "XYZ": {"Perimeter": "BCD", "Status": "Active", "RDB" :
"2345", "Zone": "USD"},
    "FX": {"Perimeter": "KJH", "Status": "Active", "RDB" :
"3456", "Zone": "HK"},
    "SNI": {"Perimeter": "BCD", "Status": "Not
Active", "RDB" : "4567", "Zone": "EUR"},
    "J3": {"Perimeter": "KJH", "Status": "Active", "RDB" :
"5678", "Zone": "EUR"}
}
ptf_data = [
{
    "PDS": [
        {"Portfolio": "PDS_ABC", "Status": "Active"},
        {"Portfolio": "PDS_XYZ", "Status": "Active"},
        {"Portfolio": "PDS_DEF", "Status": "Active"},
        {"Portfolio": "PDS_DEY", "Status": "Not Active"}
    ],
    "XYZ": [
        {"Portfolio": "XYZ_ABC", "Status": "Active"}
    ],
    "FX" : [
        {"Portfolio": "FX-GY-NOMGT", "Status": "Active"},
        {"Portfolio": "GY-MGT", "Status": "Not Active"}
    ],
    "SNI" : [
        {"Portfolio": "SN-FX-GY", "Status": "Not Active"}
    ],
    "J3" : [
        {"Portfolio": "JB-KJ-LI", "Status": "Active"}
    ]
}
]
```

1. **Checking if a GOP is active**

- **Logic:** You can find status of the GOP in `gop\_data` and returns a response.

- **Example:** If the user asks, "Is PDS GOP active?", LangChain can check the `gop\_data` dictionary and respond.

2. ****Profiles with Access to a GOP****

- ****Logic:****
 - Find the perimeter of the GOP in `gop\_data`.
 - Match that perimeter with `gop\_perimeter` to identify the profiles.
- ****Example Question:**** "Which profiles have access to FX GOP?"
 - LangChain can determine:
 - FX → KJH (from `gop\_data`), LangChain can check the `gop\_data` dictionary and respond.
 - KJH is available under the `IT` profile in `gop\_perimeter`, LangChain can check the `gop\_perimeter` dictionary and respond.
 - Response: "The IT profile has access to FX GOP."

3. ****Handling Access Error Messages****

- ****Error Context:**** LangChain can use structured prompts or retrieval-based QA to explain error messages, using your predefined explanation logic.
- ****Example Error:**** "You aren't allowed to do select with portfolio = PDS_ABC."
 - ****LangChain Output:****
 - "This error occurs because PDS_ABC is linked to PDS GOP, which is under the **BCD** perimeter. Your profile doesn't have access to this perimeter. Please check with your manager and the securities team for approval to add this gop to your profile. Once it is approved Eliot team will add the gop in Perimeter screen. Access can be given on gop level not on portfolio."